

QUBO Formulation of Low-Thrust Cislunar Trajectory Scheduling

Gino Luciano Moretta^{*1} and Antonio Paolozzi¹

¹School of Aerospace Engineering, Sapienza Università di Roma

June 2026

Abstract

We formulate low-thrust cislunar trajectory correction in the planar circular restricted three-body problem (CR3BP) as a quadratic unconstrained binary optimisation (QUBO) problem in which the decision variables are inherently binary engine on/off commands. This avoids the artificial quantisation error that arises when continuous pseudospectral trajectory coefficients are encoded into binary digits, complementing the continuous-coefficient QUBO transcription of De Grossi, Carbone et al. (2025). A first-order state-transition-matrix expansion around a reference trajectory yields a single-pass QUBO whose global minimum is reachable by exact classical (mixed-integer linear programming, brute force) and heuristic (simulated annealing) solvers, and is also expressible on D-Wave hardware. Because the linearisation is loose for long arcs, we introduce a sequential re-linearisation outer loop with a trust-region accept-or-reject step; the binary analogue of the continuous-side sequential linearisation in Carbone et al. We also report a classical MILP baseline obtained by McCormick lifting, certified optimal up to $N \approx 20$ decision steps, and a measured scaling sweep on the cislunar QUBO at N up to 200 that shows the **dwave-hybrid** Kerberos workflow finds lower-energy solutions than raw simulated annealing in $20\text{--}30\times$ less wall time at $N \geq 100$.

A *sliding-window* extension chains short single-arc QUBOs across many lunar revolutions, with an action-radius gate that suppresses scheduling far from the Moon and a velocity-only circular target that supplies a uniform braking signal in any orbital phase. The four mission phases (Earth ascent, cislunar correction, lunar capture, on-orbit station-keeping) are flown by a single 100 kg microsatellite using the native throttling envelope of one ion/Hall-class thruster with 350 mN main for the spiral and the capture and 5.4 mN for cruise and corrections. We demonstrate Phases $1 \rightarrow 2 \rightarrow 3$ composed end-to-end: $\Delta v_{\text{total}} \approx 2502 \text{ m s}^{-1}$ in $\sim 32 \text{ d}$.

Keywords: quantum annealing, QUBO, low-thrust trajectory optimisation, cislunar transfer, CR3BP, sequential linearisation, trust region.

1 Introduction

The renewed interest in cislunar logistics has sharpened the demand for low-thrust trajectory design tools that scale to many revolutions and tight timing constraints. Direct collocation and indirect Pontryagin methods are mature but rely on continuous-variable nonlinear programming, whose solver effort grows quickly with the number of discretisation nodes and the complexity of the dynamical model. Quantum annealing (QA) [8] and the quadratic-unconstrained-binary-optimisation (QUBO) formalism [7] have recently been applied to interplanetary trajectory transcription by De Grossi, Carbone et al. [1], who use the *pseudospectral* method (Lagrange-polynomial interpolation of state and control over the time interval) to discretise the continuous-time optimal control problem into a finite-dimensional NLP, linearise the dynamics about a

^{*}morettaginoluciano@gmail.com

reference trajectory, encode the resulting real-valued coefficients as binary digits, and submit the resulting QUBO to D-Wave hybrid solvers and quantum annealers. The procedure is wrapped in a sequential outer loop, in which the solution from one iteration becomes the reference for the next. Their Earth–Mars low-thrust results demonstrate that QA can reach competitive solutions on a representative aerospace optimisation problem.

The Carbone et al. formulation encodes the real-valued pseudospectral coefficients as binary digits. Direct extension to the Earth–Moon CR3BP, however, raises a question of fit between formulation and problem: the inner linear-residual step is itself solved exactly by classical methods in milliseconds, while the binary-encoded QUBO is constrained by its bit precision and approaches the same answer to within $O(2^{-p})$ per coefficient. On the Earth–Moon problem this discretisation interacts with the nonlinear-residual outer loop in ways that limit the achievable final-state accuracy. The continuous-coefficient pseudospectral QUBO and the formulation we present here are therefore best viewed as complementary, addressing different aspects of low-thrust QA.

This paper develops the second case. We pose a problem that is *naturally binary*: at each of N time steps along a reference trajectory, an on/off command $q_i \in \{0, 1\}$ decides whether a constant-thrust engine fires. There is no quantisation error because the variables are not approximations of a continuum. The combinatorial cardinality 2^N grows exactly as quantum annealing is designed to handle, and the resulting QUBO maps directly onto D-Wave hardware: N logical qubits, no embedding overhead beyond hardware connectivity.

The first-order state-transition-matrix (STM) expansion around the coast arc gives a single-pass QUBO formulation that is faithful in the small-time regime but loses accuracy as the arc lengthens or the thrust grows. Our principal methodological contribution is an outer loop that re-linearises around the current schedule and applies a trust-region accept-or-reject test using the true nonlinear miss. The loop is monotone, terminates at a fixed point, and never returns a schedule with a larger miss than the incumbent — in particular, it falls back to coast if the linearisation is so loose that the QUBO’s first candidate is non-improving (Proposition 1).

Contributions. The paper makes the following contributions:

1. A naturally-binary QUBO formulation of low-thrust cislunar trajectory scheduling in the planar CR3BP (Sec. 3.1 and 3.2).
2. A sequential re-linearisation outer loop with a trust-region accept-or-reject test, monotone in the true nonlinear miss (Sec. 3.3).
3. A classical mixed-integer linear-programming (MILP) baseline obtained by the McCormick lift of the binary-quadratic objective, providing certified optimality at small to medium N and an honest scaling reference against simulated annealing (Sec. 4).
4. A sliding-window extension of the iterative driver that chains short single-arc QUBOs across many lunar revolutions, with a window-length cap and an action-radius gate that confine scheduling to the regime where the Moon-relative linearisation is faithful (Sec. 3.4).
5. Validation of simulated annealing against brute force, a single-pass-vs-iterative trajectory comparison at long T , a second application of the same method to lunar-orbit station-keeping, a head-to-head comparison with continuous direct-collocation NLP, and a scaling sweep over brute force, MILP, simulated annealing, and the dwave-hybrid Kerberos workflow up to $N = 200$ qubits (Sec. 5).
6. An end-to-end Earth–Moon mission demonstration in which every Δv is delivered by binary on/off commands of a single throttleable ion/Hall thruster: the Phase-1 spiral absorbs the cislunar injection burn, Phase 2 corrects the flyby at cruise power, and the sliding-window QUBO captures from the resulting hyperbolic flyby into a near-circular bound lunar orbit at main power, with no impulsive burns (Sec. 5.9).

2 Problem statement

2.1 Planar circular restricted three-body problem

Working in the planar synodic (rotating) frame [2, 3], with the standard non-dimensionalisation that places Earth at $(-\mu, 0)$ and the Moon at $(1-\mu, 0)$ with mass parameter $\mu = m_{\text{Moon}}/(m_{\text{Earth}} + m_{\text{Moon}}) \approx 0.01215$, the equations of motion under thrust acceleration $\mathbf{u}(t) \in \mathbb{R}^2$ are

$$\begin{aligned}\ddot{x} - 2\dot{y} &= \Omega_x(x, y) + u_x, \\ \ddot{y} + 2\dot{x} &= \Omega_y(x, y) + u_y,\end{aligned}\tag{1}$$

with the synodic pseudo-potential $\Omega(x, y) = \frac{1}{2}(x^2 + y^2) + (1 - \mu)/r_1 + \mu/r_2 + \frac{1}{2}\mu(1 - \mu)$ and primary distances r_1, r_2 . The state is $\mathbf{x} = [x, y, \dot{x}, \dot{y}]^\top \in \mathbb{R}^4$ and the control influence matrix $\mathbf{B} = [\mathbf{0}_{2 \times 2}; \mathbf{I}_2] \in \mathbb{R}^{4 \times 2}$ is constant. We use the Earth–Moon nondimensionalisation $L \approx 384\,400$ km, $T \approx 4.343$ days, $V \approx 1023$ m/s.

2.2 Scheduling decision space

A reference initial state $\mathbf{x}_0$, a target final state $\mathbf{x}_{\text{target}}$, and a time interval $[t_0, t_f]$ are given. The interval is divided into N equal decision steps of duration $\Delta t = (t_f - t_0)/N$. The thrust at step i is either zero (coast, $q_i = 0$) or a constant vector $\mathbf{u}_i \in \mathbb{R}^2$ in a prescribed direction (burn, $q_i = 1$); typical direction modes are *tangential* ($\hat{\mathbf{u}} = \dot{\mathbf{x}}/\|\dot{\mathbf{x}}\|$), its opposite, the velocity-normal directions, or a fixed inertial direction. In the multi-channel variant, K independent direction channels each carry their own per-step binary, giving $M = KN$ binary decisions $\mathbf{q} = (q_{1,1}, \dots, q_{1,N}, q_{2,1}, \dots, q_{2,N}, \dots) \in \{0, 1\}^M$. The optimisation problem is

$$\min_{\mathbf{q} \in \{0,1\}^M} \|\mathbf{x}(t_f; \mathbf{q}) - \mathbf{x}_{\text{target}}\|_{\mathbf{W}}^2 + \alpha \|\mathbf{q}\|_1 + (\text{exclusion penalty}),\tag{2}$$

where $\mathbf{x}(t_f; \mathbf{q})$ is the final state of the nonlinear CR3BP propagated under the schedule $\mathbf{q}$, $\mathbf{W} \succeq 0$ weights the state components, and $\alpha \geq 0$ is a fuel-cost multiplier. The exclusion penalty enforces single-engine operation when needed (Sec. 3.2).

3 Method

3.1 Single-pass STM linearisation

Let $\mathbf{x}_{\text{coast}}(t)$ denote the unforced trajectory from $\mathbf{x}_0$ at time t_0 and let $\Phi_{\text{coast}}(t, t_0)$ be the corresponding state-transition matrix, satisfying $\dot{\Phi} = (\partial f / \partial \mathbf{x})|_{\mathbf{x}_{\text{coast}}(t)} \Phi$, $\Phi(t_0, t_0) = \mathbf{I}$. To first order in the control, the impulse $\mathbf{u}_i \Delta t$ applied at t_i shifts the final state by $\Phi(t_i, t_f) \mathbf{B} \mathbf{u}_i \Delta t$, where $\Phi(t_i, t_f) = \Phi(t_f, t_0) \Phi(t_0, t_i)$. Defining the per-variable *impulse response vector*

$$\mathbf{b}_j = \Phi(t_i, t_f) \mathbf{B} \mathbf{u}_{c,i} \Delta t, \quad j = cN + i,\tag{3}$$

where $\mathbf{u}_{c,i}$ is the thrust vector for channel c at step i (e.g. tangential evaluated along the coast arc), the linearised final state is

$$\mathbf{x}(t_f; \mathbf{q}) \approx \mathbf{x}_{\text{coast}}(t_f) + \sum_{j=1}^M q_j \mathbf{b}_j.\tag{4}$$

Substituting into (2), with effective state gap $\mathbf{d} = \mathbf{x}_{\text{target}} - \mathbf{x}_{\text{coast}}(t_f)$, gives a quadratic-in- $\mathbf{q}$ objective

$$E(\mathbf{q}) = \left\| \sum_j q_j \mathbf{b}_j - \mathbf{d} \right\|_{\mathbf{W}}^2 + \alpha \sum_j q_j = \mathbf{q}^\top \mathbf{Q} \mathbf{q} + \boldsymbol{\ell}^\top \mathbf{q} + c_0,\tag{5}$$

which is a QUBO. The matrix entries follow directly from expanding the squared norm: $Q_{jk} = \mathbf{b}_j^\top \mathbf{W} \mathbf{b}_k$, $\ell_j = \alpha - 2\mathbf{b}_j^\top \mathbf{W} \mathbf{d}$, $c_0 = \mathbf{d}^\top \mathbf{W} \mathbf{d}$. For binary variables $q_j^2 = q_j$, so the diagonal of $\mathbf{Q}$ is absorbed into $\boldsymbol{\ell}$ when posting to a sampler; the QUBO has M logical qubits and dense couplings.

Number of qubits. For a single-channel problem with N decision steps the QUBO has $M = N$ qubits; for a K -channel multi-direction problem $M = KN$. Both are linear in the discretisation, in contrast to the quadratic growth of pseudospectral-coefficient QUBOs whose qubit count scales as $p \cdot L \cdot (\text{state dim})$ for p bits per coefficient and L basis functions.

3.2 Multi-channel scheduling and exclusion

The multi-channel QUBO assembles directly from (3)–(5) with the variable index $j = cN + i$ running over all (c, i) pairs. Channel directions are evaluated at the reference-trajectory state at t_i , so they follow the spacecraft’s velocity smoothly along the arc.

When the spacecraft has a single engine that cannot fire two directions simultaneously, an *exclusion penalty*

$$E_{\text{excl}}(\mathbf{q}) = P \sum_{i=1}^N \sum_{c < c'} q_{c,i} q_{c',i} \quad (6)$$

is added to $E(\mathbf{q})$. The penalty P is chosen large enough that any pair of simultaneous burns at the same step costs more than the best feasible alternative; in practice P on the order of the largest $|Q_{jk}|$ suffices. The penalty couples same-step variables across channels but adds no auxiliary variables, preserving the QUBO structure.

3.3 Iterative re-linearisation

The single-pass formulation (4) is exact only in the small-time / small-thrust limit. For arcs in which the spacecraft moves significantly through configuration space, or where strong gravitational gradients make $\partial f / \partial \mathbf{x}$ vary along the trajectory, the linearisation error is non-negligible and the QUBO objective is no longer a faithful proxy for the true nonlinear miss.

We address this by re-linearising around the current schedule in an outer loop. Let $\mathbf{q}^{(k)}$ be the schedule at iteration k , and let $\mathbf{x}_{\text{nom}}^{(k)}(t)$ be the trajectory obtained by propagating the true nonlinear CR3BP under the controls implied by $\mathbf{q}^{(k)}$. We linearise the final state about this nominal:

$$\mathbf{x}(t_f; \mathbf{q}) \approx \mathbf{x}_{\text{nom}}^{(k)}(t_f) + \sum_j (q_j - q_j^{(k)}) \mathbf{b}_j^{(k)}, \quad (7)$$

where $\mathbf{b}_j^{(k)}$ is built from (3) using STMs along $\mathbf{x}_{\text{nom}}^{(k)}(t)$. Substituting into (2) and absorbing the linear shift into an *effective* target gap

$$\mathbf{d}_{\text{eff}}^{(k)} = (\mathbf{x}_{\text{target}} - \mathbf{x}_{\text{nom}}^{(k)}(t_f)) + \sum_j q_j^{(k)} \mathbf{b}_j^{(k)}, \quad (8)$$

yields a QUBO of the same algebraic form as (5) with $\mathbf{d} \rightarrow \mathbf{d}_{\text{eff}}^{(k)}$ and $\mathbf{b}_j \rightarrow \mathbf{b}_j^{(k)}$. The variable space is unchanged: $\mathbf{q} \in \{0, 1\}^M$ remains the absolute schedule, not a perturbation.

The outer loop. The iterative driver is Algorithm 1.

Monotonicity and the trust-region rule. Step 2(e) is the key safeguard. The linearised model (7) is only locally accurate: a candidate $\hat{\mathbf{q}}$ that the QUBO objective ranks below the incumbent may actually have a larger true nonlinear miss. The accept-or-reject rule converts this risk into a formal guarantee.

Proposition 1. Let $\{\mathbf{q}^{(k)}\}_{k=0}^K$ be the sequence of accepted schedules produced by Algorithm 1 with initial schedule $\mathbf{q}^{(0)} = \mathbf{0}$, and let $r^{(k)} = \|\mathbf{x}(t_f; \mathbf{q}^{(k)}) - \mathbf{x}_{\text{target}}\|$ be the corresponding true nonlinear miss. Then

(i) the sequence $r^{(k)}$ is non-increasing, $r^{(0)} \geq r^{(1)} \geq \dots \geq r^{(K)}$;

Algorithm 1. Iterative re-linearisation (`solve_iterative`).

1. Initialise $\mathbf{q}^{(0)} = \mathbf{0}$ (coast). Compute the incumbent miss $r^{(0)} = \|\mathbf{x}(t_f; \mathbf{0}) - \mathbf{x}_{\text{target}}\|$.
2. For $k = 0, 1, 2, \dots, k_{\text{max}} - 1$:
 - (a) Build the QUBO linearised around $\mathbf{q}^{(k)}$ using (7)–(8).
 - (b) Sample a candidate $\hat{\mathbf{q}}$ from the QUBO via any sampler (brute force, MILP, simulated annealing, Leap Hybrid, D-Wave QPU).
 - (c) If $\hat{\mathbf{q}} = \mathbf{q}^{(k)}$: terminate (fixed point).
 - (d) Propagate the nonlinear CR3BP under $\hat{\mathbf{q}}$ and compute the candidate miss $\hat{r} = \|\mathbf{x}(t_f; \hat{\mathbf{q}}) - \mathbf{x}_{\text{target}}\|$.
 - (e) *Trust-region accept*: if $\hat{r} > r^{(k)}$, terminate (return $\mathbf{q}^{(k)}$).
 - (f) Set $\mathbf{q}^{(k+1)} = \hat{\mathbf{q}}$ and $r^{(k+1)} = \hat{r}$.
 - (g) If $r^{(k)} - r^{(k+1)} < \varepsilon_{\text{tol}}$: terminate.
3. Return $\mathbf{q}^{(K)}$, $r^{(K)}$, and the iteration history.

(ii) the algorithm terminates in finitely many outer iterations, $K \leq k_{\text{max}}$;

(iii) the returned schedule is at least as good as coast: $r^{(K)} \leq r^{(0)} = \|\mathbf{x}_{\text{coast}}(t_f) - \mathbf{x}_{\text{target}}\|$.

Proof. A schedule $\hat{\mathbf{q}}$ is admitted as $\mathbf{q}^{(k+1)}$ at step 2(f) of Algorithm 1 only if step 2(e) does not terminate, i.e. if $\hat{r} \leq r^{(k)}$. Setting $r^{(k+1)} = \hat{r}$ gives $r^{(k+1)} \leq r^{(k)}$, which establishes (i) by induction. The algorithm exits the for-loop in step 2 either at iteration k_{max} or earlier, and any earlier exit happens through steps 2(c), 2(e), or 2(g), giving (ii). Property (iii) follows from (i) and the choice $\mathbf{q}^{(0)} = \mathbf{0}$, for which $\mathbf{x}(t_f; \mathbf{q}^{(0)}) = \mathbf{x}_{\text{coast}}(t_f)$. $\square$

The contrast with the unguarded single-pass solve is sharp: the single-pass minimiser of the linearised QUBO can have a true nonlinear miss *larger* than coast (see Sec. 5.3, Fig. 2, $T = 5$ case), whereas Algorithm 1 by Proposition 1(iii) cannot.

Cost of the loop. Each outer iteration requires (i) one re-propagation of the nonlinear dynamics with state and state-transition-matrix integration to assemble $\mathbf{b}_j^{(k)}$, costing $O(N \cdot n_{\text{sub}})$ RK4 steps, and (ii) one sampler call. With brute force, simulated annealing, or MILP at $N = 15$ each iteration takes a few seconds on a laptop; on a QPU each iteration is dominated by the cloud round-trip ($\sim$ seconds) since the actual anneal is microseconds. The experiments in Sec. 5 show that 1–4 outer iterations suffice across the regime where the method is applicable.

3.4 Sliding-window QUBO for lunar capture

The iterative re-linearisation of Sec. 3.3 closes the linearisation gap of a *single* arc but does not by itself extend to multi-revolution manoeuvres. A representative case is lunar orbit insertion (LOI) from a hyperbolic flyby: the spacecraft passes the Moon once, must shed several tens of metres per second of excess velocity to bind its orbit, and then circularise over many subsequent revolutions. A single-arc QUBO covering this entire process would have a linearisation gap so large that no schedule survives the trust-region rule of Algorithm 1.

We address this with a *sliding-window* structure. The mission is split into W short arcs of duration ΔT_w , each of which is small enough that the iterative re-linearisation of Sec. 3.3 converges. Between windows the spacecraft state is propagated under the true nonlinear CR3BP and becomes the initial state of the next window. The schedule is the concatenation of per-window schedules.

Window-length capping. The natural choice for ΔT_w is a fraction of the local Moon-relative orbital period $T_{\text{loc}} = 2\pi\sqrt{r_w^3/\mu}$, with r_w the spacecraft’s Moon-relative radius at the start of window w . This adapts the window to the local dynamics. However, T_{loc} diverges as r_w grows (loose orbits or escapes), and an unguarded fractional-period window would become arbitrarily long once the orbit becomes large – by which point the linearisation has long since collapsed. We therefore cap ΔT_w at a fixed ΔT_{max} so that the linearisation regime remains valid even when the orbit is loose.

Action-radius gating. A second safeguard restricts QUBO scheduling to windows in which the spacecraft is within a threshold r_{act} of the Moon. Outside this radius, the dynamics are dominated by Earth’s gravity and the Moon-relative linear target is meaningless: every candidate fails the trust-region rule and the loop produces no progress while burning through windows. With gating, those windows become unforced drifts – the spacecraft is propagated freely under the CR3BP until it returns inside r_{act} , at which point QUBO scheduling resumes. Choosing r_{act} on the order of a few perilune radii confines the QUBO action to the region where anti-tangential thrust efficiently dissipates orbital energy.

Per-window target. For each window we set the QUBO target to a state whose Moon-relative inertial velocity has magnitude $v_{\text{circ}}(r) = \sqrt{\mu/r}$ in the direction of the spacecraft’s current Moon-relative velocity, evaluated at the unforced end-of-window position $\mathbf{x}_{\text{eow}}$:

$$\mathbf{x}_{\text{target}}^{(w)} = [\mathbf{r}_{\text{eow}}, v_{\text{circ}}(\|\mathbf{r}_{\text{eow}} - \mathbf{r}_{\text{Moon}}\|) \hat{\mathbf{v}}_{\text{Moon-rel}}^{\text{eow}}] \quad (9)$$

expressed in the synodic frame after the standard $\boldsymbol{\omega} \times \mathbf{r}$ correction. Position is left unweighted ($W_{\text{pos}} = 0$) and only velocity components are weighted in the QUBO objective: the target says “stay where the unforced trajectory takes you, but slow to the local circular speed”. With a single anti-tangential channel, this target elicits firings precisely where the spacecraft is moving faster than circular, which is the energy-dissipation regime for lunar capture and circularisation. At apolune the spacecraft is already slower than v_{circ} , anti-tangential thrust cannot speed it up, and the QUBO correctly stays off.

Termination and diagnostics. The outer loop terminates after W windows have been processed. We monitor the two-body energy $E_w = \frac{1}{2}\|\mathbf{v}_w^{\text{rel}}\|^2 - \mu/\|\mathbf{r}_w^{\text{rel}}\|$ and the Moon-relative apolune radius r_{apo} across windows. Captured states have $E_w < 0$ and finite r_{apo} ; loose-orbit windows have r_{apo} in the tens of thousands of kilometres; escaped or pre-capture states have $E_w \geq 0$ and $r_{\text{apo}} = \infty$.

Per-window guarantees. Each window is itself a call to Algorithm 1 with a short ΔT_w , so Proposition 1 applies window-by-window: the schedule produced for window w has true nonlinear miss no worse than coast *within that window*. The composition does not admit a global monotonicity guarantee in E_w because the unforced drift between windows can both raise or lower the energy under CR3BP perturbations, but in practice anti-tangential firings near perilune dominate and the energy decreases over the course of the schedule (see Sec. 5.9).

4 Classical baseline: MILP via McCormick lift

The QUBO (5) can be solved exactly by classical mixed integer linear programming. Introduce auxiliary binaries y_{ij} for $i < j$ and the McCormick constraints [9]

$$y_{ij} \leq q_i, \quad y_{ij} \leq q_j, \quad y_{ij} \geq q_i + q_j - 1, \quad y_{ij} \in \{0, 1\}. \quad (10)$$

For binary q_i, q_j , (10) forces $y_{ij} = q_i q_j$ at every feasible integer point. The QUBO energy then becomes

$$E(\mathbf{q}, \mathbf{y}) = \sum_i (Q_{ii} + \ell_i) q_i + \sum_{i < j} 2 Q_{ij} y_{ij} + c_0, \quad (11)$$

which is linear in $(\mathbf{q}, \mathbf{y}) \in \{0, 1\}^{M+M(M-1)/2}$. We solve (11) with the open-source HiGHS solver [10] through `scipy.optimize.milp`, which provides a primal-dual optimality certificate at termination.

The McCormick lift adds $M(M-1)/2$ auxiliary binaries and $3M(M-1)/2$ linear constraints. Because the impulse-response vectors $\mathbf{b}_j \in \mathbb{R}^4$ couple every pair, $\mathbf{Q}$ is dense and the linear-programming relaxation of (11) is loose. In practice this means HiGHS certifies optimality quickly up to $N \approx 20$ and struggles beyond, which is sufficient as a small- N ground truth for benchmarking but is not a competitive heuristic at large scale.

5 Experiments

5.1 Mission scenario and spacecraft

The experiments in this section all describe the same Earth–Moon mission, flown by a single spacecraft, demonstrated phase by phase and then composed end-to-end. The spacecraft is a ~ 100 kg microsatellite with a single throttleable ion/Hall-class propulsion system. The same physical thruster delivers different effective accelerations depending on the input power. We exploit this native throttling and operate the thruster at two of its native points:

- *Main* power: 350 mN, $a_{\text{main}} \approx 3.5 \text{ mm s}^{-2}$ ($a = 1.286$ in CR3BP nondim units).
- *Cruise* power: 5.4 mN, $a_{\text{cruise}} \approx 0.054 \text{ mm s}^{-2}$ ($a = 0.02$ in CR3BP nondim units).

Throttling is intrinsic to the propulsion system; switching power levels does not require additional hardware. The choice of mode in each mission phase is dictated by the QUBO discretisation requirement that one binary slot’s Δv be comparable to the trajectory increment of that phase.

The mission has four phases:

- Phase 1** Earth ascent: an analytical Edelbaum spiral [6] from a high parking orbit ($r \approx 42\,164$ km, GEO altitude) to a $r = 0.5$ nondim ($\sim 200\,000$ km) orbit, followed by a continuous tangential boost into a Moon-bound trajectory. Both segments run at *main* power as a single all-on binary schedule.
- Phase 2** Cislunar correction: the QUBO scheduler of Sec. 3.3 corrects the imperfect injection at *cruise* power, where each binary slot delivers $\sim 4.4 \text{ m s}^{-1}$.
- Phase 3** Lunar capture and stabilisation: the sliding-window scheduler of Sec. 3.4 fires the thruster at *main* power during each perilune passage, dissipating the hyperbolic excess energy across several short windows until the orbit becomes bound and near-circular.
- Phase 4** On-orbit station-keeping: once on a stable lunar orbit, the iterative driver of Sec. 3.3 maintains the orbit against CR3BP perturbations at *cruise* power, position-only target weighting.

The remainder of Sec. 5 demonstrates each phase in isolation as a focused experiment (Sec. 5.3 for Phase 2, Sec. 5.5 for Phase 4) and then composes Phases 1, 2, and 3 into a single end-to-end transfer (Sec. 5.9). The cislunar injection-correction sub-scenario used throughout Sec. 5.3–5.6–5.7 is the Phase 2 segment of this mission, run in cruise mode. Figure 1 shows the resulting trajectory for $v_{\text{inj}}/v_{\text{circ}} = 1.188$, $T = 4.0$ nondim (~ 17.4 days), $N = 20$.

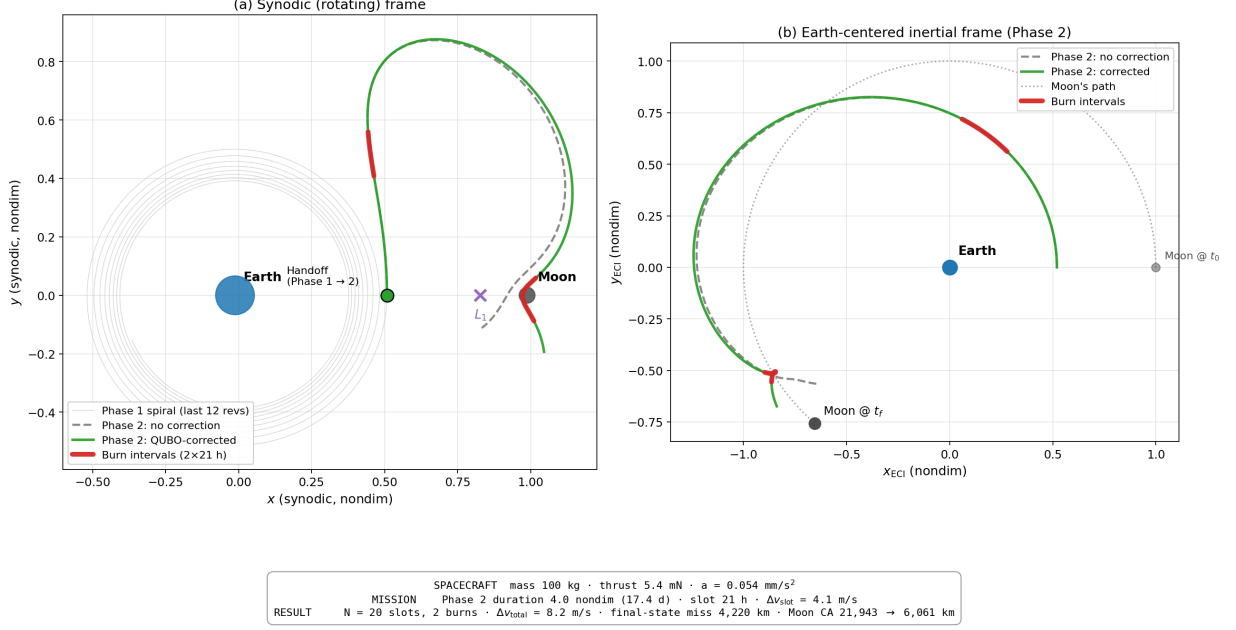


Figure 1: Phase 2 of the mission: cislunar correction trajectory in two reference frames. (a) CR3BP synodic (rotating) frame: the last twelve revolutions of the Phase-1 Edelbaum spiral are visible as light-gray ellipses arriving at the green Handoff marker; from there the Phase-2 coast (gray dashed) and QUBO-corrected (green) trajectories diverge toward the Moon, with the two active burn intervals shown as thick red segments along the corrected trajectory. (b) Earth-centered inertial frame, Phase-2 only: the same trajectories appear as a curving arc over ~ 17 days, with the Moon advancing along its circular path from t_0 to t_f . The bottom panel summarises the spacecraft, mission, and result parameters. The thruster runs in cruise mode for this phase ($a \approx 0.054 \text{ mm/s}^2$, $\sim 5.4 \text{ mN}$ out of the 350 mN envelope of the throttleable 100 kg microsatellite). The iterative QUBO selects two 21-hour burns for a total Δv of 8.2 m/s, reducing the lunar closest approach from $\sim 22\,000 \text{ km}$ (coast) to $\sim 6\,000 \text{ km}$ (corrected).

5.2 Validation: simulated annealing matches brute force

For $N = 15$ ($2^{15} = 32\,768$ candidate schedules) and at all multi-channel sizes up to $M = 40$ qubits used in this section, we solve the QUBO with brute force and with simulated annealing (`neal.SimulatedAnnealingSampler`, 1000–4000 reads). Simulated annealing recovers the brute-force optimum on every problem size where the comparison is computable, so simulated annealing is a faithful surrogate at this size and a defensible upstream proxy for quantum-annealing performance.

5.3 Phase 2 demonstration: linearisation gap, single-pass vs iterative

This subsection focuses on Phase 2 of the mission introduced in Sec. 5.1: the cislunar trajectory correction at cruise power. To quantify the value of the iterative re-linearisation we run a long-arc cislunar correction at $T = 4$ nondim (~ 17 days), $N = 15$, $v_{\text{inj}}/v_{\text{circ}} = 1.187$. At this T the single-pass linearisation about the coast arc is loose enough that the QUBO’s optimum schedule, when propagated under the true nonlinear CR3BP, lands far from the target. Figure 2 compares the actual trajectories obtained from the two solvers on this scenario.

The figure shows the operational difference between the two methods. The single-pass schedule has the QUBO believing the linearised miss is small (142 000 km predicted $\approx 143\,000 \text{ km}$ true), but the linearisation around the unforced coast arc is so far from the actual thrust-applied dynamics at $T = 4$ that the schedule delivers essentially what coast would. The iterative loop’s first outer iteration relinearises around the single-pass schedule; subsequent iterations refine the linearisation

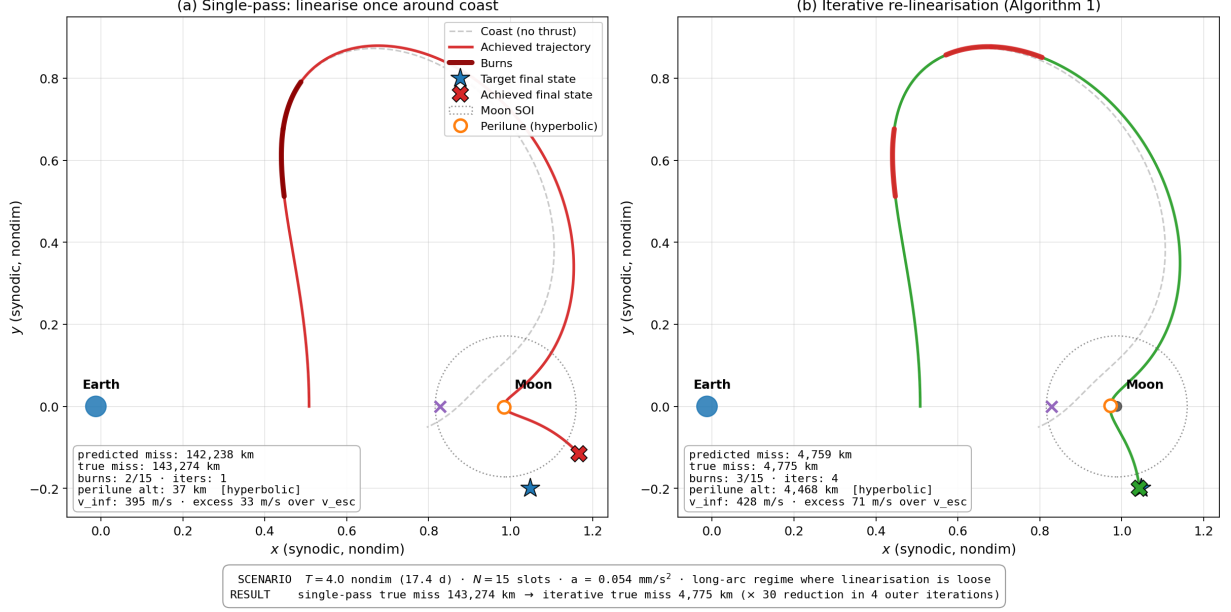


Figure 2: Single-pass QUBO (a) versus iterative re-linearisation (b) on a long-arc cislunar correction at $T = 4$ nondim. The blue star is the target final state; the cross is the achieved final state. The single-pass solver predicts a 142 000 km miss and delivers 143 000 km. Meanwhile the iterative solver, after four iterations and three burns it lands within 4800 km of the target, a 30 $\times$ reduction in true miss for the cost of one extra burn. The operational regime of Phase 2 (Sec. 5.9) lands the corrected state within ~ 1900 km of the target, inside the Moon SOI, where the Phase-3 sliding-window scheduler takes over.

about progressively closer trajectories until the QUBO’s predicted miss converges to the true nonlinear miss within numerical tolerance.

Beyond $T \gtrsim 5$ nondim (~ 22 days) the linearisation gap is so wide that no schedule satisfies the trust-region accept rule on the first iteration; the loop correctly terminates at coast.

5.4 Connection to primer-vector theory

The on/off scheduling QUBO never computes a Pontryagin costate. Once a schedule has converged under the iterative driver, however, a costate can be reconstructed *a posteriori* along the resulting nonlinear trajectory by integrating $\dot{\boldsymbol{\lambda}} = -\mathbf{A}(\mathbf{x}(t))^{\top} \boldsymbol{\lambda}$ backward in time from the terminal transversality condition $\boldsymbol{\lambda}(t_f) = 2\mathbf{W}(\mathbf{x}_f - \mathbf{x}_{\text{target}})$. The tangential primer-vector analogue is the scalar switching function

$$S(t) = -\boldsymbol{\lambda}_v(t) \cdot \hat{\mathbf{v}}(t), \quad (12)$$

where $\boldsymbol{\lambda}_v$ is the velocity sub-vector of the costate and $\hat{\mathbf{v}}$ the unit velocity vector. Pontryagin’s minimum principle predicts that the optimal tangential bang-bang control fires the engine when $S > 0$ and coasts otherwise. Comparing this continuous-time prediction with the QUBO’s discrete schedule provides a sanity check on the bridge between the binary formulation and indirect primer-vector methods [13].

Applied to the converged iterative schedule of Sec. 5.3 ($T = 4$ nondim, $N = 15$, three ON slots out of fifteen), the reconstruction yields a switching function $S(t)$ whose support is a contiguous interval near the second half of the arc.

This reconstruction is, strictly speaking, a self-consistency check: $\boldsymbol{\lambda}$ is built from the QUBO’s own final state and propagated through the QUBO’s own trajectory. A fully external comparison would solve the same problem via indirect methods (the route taken in [13] via homotopy continuation in ρ and u_{max}) and overlay the two switching functions. We leave that comparison

to future work, as the in-house direct-collocation NLP of Sec. 5.6 is energy-optimal rather than fuel-optimal and therefore lacks a directly comparable primer vector.

5.5 Phase 4 demonstration: lunar-orbit station-keeping

This subsection demonstrates Phase 4 of the mission of Sec. 5.1: long-duration station-keeping at the operational lunar orbit, run at cruise power (5.4 mN, the same operating point as Phase 2 in Sec. 5.3). The scenario is a 5 000-km circular lunar orbit perturbed by a 0.5% tangential velocity error at t_0 , with the iterative QUBO scheduling a single anti-tangential channel to maintain the orbit against this perturbation. Over two revolutions (~ 28 hours) the unperturbed coast trajectory drifts by about 1 300 km in position from the nominal endpoint; the QUBO’s 4.4 m/s schedule reduces the position miss to ~ 530 km, a $2.4\times$ improvement. Both the coast (gray) and corrected (green) tracks remain near-circular about the Moon when plotted in the Moon-relative frame, with the burn intervals appearing as short red segments distributed across the two revolutions rather than concentrated at a single anomaly.

The QUBO uses the position-only target-weighting variant of (2): a phase shift across the orbit affects the velocity vector but not the position objective, so a station-keeping mission that cares about position benefits from the diagonal weight $W = \text{diag}(1, 1, 0, 0)$. This is a different application of the same method as Sec. 5.3, with no changes to the QUBO formulation or the iterative driver.

This scenario is in the linearisation-faithful regime so the iterative loop terminates after one outer iteration with the schedule being a fixed point. The contribution here is not the iteration depth but the formulation: an on/off binary scheduling problem on a QUBO solver naturally accommodates the pulse-mode operation of real spacecraft propulsion, while the trust-region rule keeps the method robust across thrust regimes.

5.6 Comparison with continuous-control direct collocation

The standard aerospace-engineering tool for low-thrust trajectory optimisation is direct collocation NLP — typically Hermite-Simpson or Legendre–Gauss–Lobatto with SLSQP / IPOPT [4, 5]. To position our QUBO methodology against that benchmark we re-pose the cislunar correction as an energy-optimal continuous-control problem with boundary conditions enforced as equalities, and solve it with an in-house Hermite-Simpson direct-collocation implementation (40 intervals, SLSQP, $\text{tol} = 10^{-9}$).

Scenario	Method	Miss (km)	Δv (m/s)	Solve (s)
$v_{\text{inj}}/v_{\text{circ}} = 1.187, T = 2$	QUBO iter. ($N=15$)	2 053	10.9	3.1
	NLP HS-direct	0.0	7.7	1.5
$v_{\text{inj}}/v_{\text{circ}} = 1.187, T = 3$	QUBO iter. ($N=15$)	1 087	8.2	4.6
	NLP HS-direct	0.0	8.8	3.2
$v_{\text{inj}}/v_{\text{circ}} = 1.185, T = 3$	QUBO iter. ($N=15$)	2 626	20.5	3.4
	NLP HS-direct	0.0	14.6	1.6

Table 1: Iterative QUBO scheduling vs continuous-control direct collocation NLP on three cislunar correction scenarios. The NLP enforces final state as an equality constraint and reaches the target exactly; the QUBO is restricted to a binary tangential action set and leaves a residual miss of 1 000–2 600 km. The Δv cost of the QUBO is comparable but slightly higher in the strong-error case; solve times are within a factor of two.

The NLP wins on the standard metric: zero miss at comparable or lower Δv . This is expected (it has a richer action space and explicit BC satisfaction). The QUBO methodology developed

here is appropriate when the action space is genuinely discrete, a regime in which the classical NLP is itself not directly applicable without post-processing.

The continuous “optimum” is solver- and seed-dependent. A caveat tempers the apparent NLP advantage of Table 1: the Hermite–Simpson transcription of the energy-optimal CR3BP transfer is a *nonconvex* NLP, so the “classical optimum” it returns is not unique — it depends on the starting iterate and on the optimiser. We make this explicit with a multistart study. For three boundary-condition cases we draw 32 random seeds (Gaussian perturbations of the linear-interpolation guess) and solve each from that seed with SLSQP; we additionally cross-check a subset of the *same* seeds with MATLAB’s `fmincon` (SQP) on the identical transcription. Figure 3 reports the result. Almost every feasible converged solution is a *distinct* local minimum (case A: 15 feasible starts, 15 distinct minima), and the objective spans more than an order of magnitude ($J \in [1.41, 19.70]$, a $\sim 1300\%$ spread). The canonical linear-interpolation seed lands at $J \approx 4.01$, well above the multistart best of $J \approx 1.41$. From identical seeds SLSQP and `fmincon` descend into different basins in five of six cases, yet agree to $\sim 10^{-9}$ whenever they share a basin — confirming that the disagreement is genuine nonconvexity of the problem, not a mismatch between the two transcriptions.

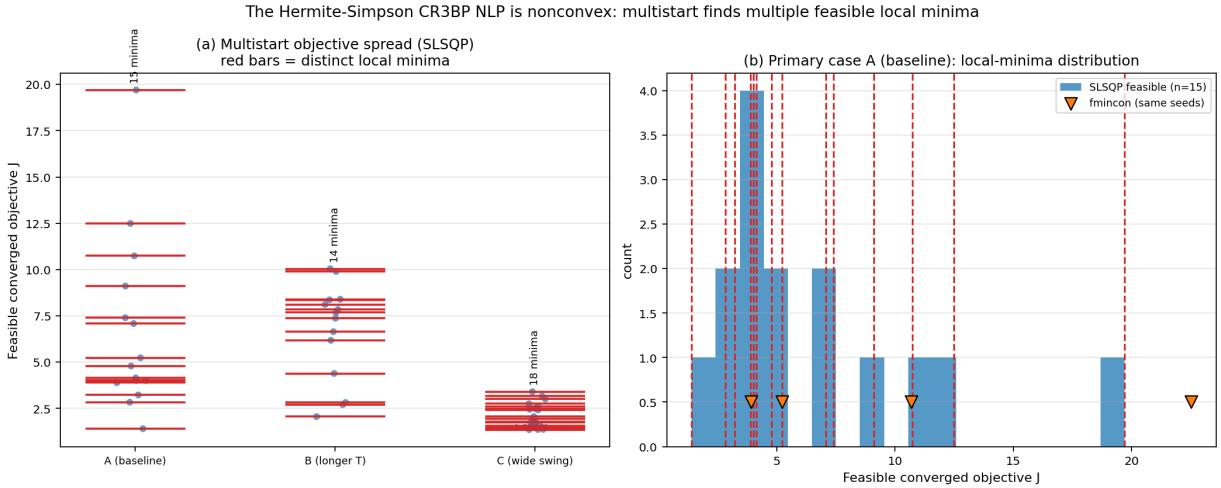


Figure 3: The Hermite–Simpson energy-optimal CR3BP transcription is nonconvex. *Left*: for three boundary-condition cases, the feasible converged objectives of 32 random multistarts (SLSQP); red bars mark distinct local minima. Nearly every feasible start reaches a different minimum, and the objective spread exceeds an order of magnitude. *Right*: primary case A, the distribution of feasible SLSQP minima (bars) with MATLAB `fmincon` results from the same seeds overlaid (orange triangles). The two optimisers scatter across the same multi-basin landscape — `fmincon` even reaches a minimum beyond the SLSQP range — but coincide to machine precision when seeded into a shared basin.

This does not overturn the comparison of Table 1, but it reframes it: quoting a single continuous optimum as “the” baseline is fragile, and any statement that the binary QUBO is within $x\%$ of optimal is only as meaningful as the seed used to compute that baseline. The exact, seed-independent reference for the binary problem remains the McCormick-lifted MILP of Sec. 4, which certifies the global binary optimum; `fmincon` corroborates the continuous transcription, and the multistart spread is itself a strong motivation for the global/combinatorial search that the QUBO formulation is built to enable.

5.7 Multi-channel scheduling

The single-channel formulation restricts thrust to one direction. Adding channels (anti-tangential, normal, anti-normal) gives the QUBO direction freedom at the cost of more variables ($M = KN$). We run the iterative scheduler on three injection-error scenarios at matched $N = 10$, $T = 3$ nondim, and report the result with single channel ($K = 1$, $M = 10$) versus four channels ($K = 4$, $M = 40$). Single-channel uses brute force; multi-channel uses simulated annealing with 4000 reads.

Scenario	Mode	Miss (km)	Δv (m/s)	Burns
Speed-only	tangential ($M=10$)	3 729	24.6	4
	4-channel ($M=40$)	880	153.5	25
Mixed	tangential ($M=10$)	2 950	18.4	3
	4-channel ($M=40$)	1 624	165.8	27

Table 2: Multi-channel vs single-channel iterative scheduling at $T = 3$ nondim, $a = 0.02$, $N = 10$. The 4-channel mode adds anti-tangential, normal, and anti-normal directions. It reduces miss by a factor of ~ 4 on the speed-error scenario and ~ 1.8 on the mixed scenario, at a substantially higher Δv .

The speed-error and mixed scenarios show the expected pattern: extra direction freedom permits substantially smaller miss when the gap is not aligned with the spacecraft velocity. The Δv cost rises roughly linearly with the number of active burns (each burn fires one engine for Δt at acceleration a), so the methodology trades fuel for accuracy in the natural way.

5.8 Scaling

We measured solve times and best energies for brute force, McCormick-lifted MILP via HiGHS, simulated annealing, and the local Kerberos workflow from **dwave-hybrid** on the cislunar scenario at increasing N . All four solvers operate on the identical QUBO. Brute force is the optimal-by-construction reference at $N \leq 20$.

Three operational regimes emerge from the data. For $N \leq 20$ all four solvers find the global optimum and selection is purely on speed (SA fastest, brute force slowest). For $20 < N \lesssim 50$ MILP can no longer certify optimality but the LP relaxation is still tight enough that the incumbent is competitive with SA. For $N \gtrsim 100$ MILP collapses (the dense $\mathbf{Q}$ makes the relaxation order-of-magnitude loose) and the choice is between the SA heuristic and the hybrid Kerberos workflow. Kerberos in this regime returns lower-energy solutions than SA in significantly less time, consistent with the design intent of dwave-hybrid, which decomposes the problem and runs classical heuristics on the parts in parallel.

The scientifically relevant claim is therefore not that quantum annealing offers an asymptotic speedup over classical brute force, but that the hybrid workflow architecture can match or beat raw simulated annealing at large N . A direct submission of the same QUBO to a Leap Hybrid solver (or to an Advantage 2 QPU when access is available) is the natural extension of this comparison and is implemented behind the sampler interface in the accompanying code.

5.9 End-to-end mission: composing Phases 1, 2 and 3

The previous subsections demonstrated each phase of the mission of Sec. 5.1 in isolation: Phase 2 cislunar correction in Sec. 5.3, Phase 4 station-keeping in Sec. 5.5. We now compose Phases 1, 2, and 3 into a single end-to-end Earth-to-bound-lunar-orbit transfer flown by the same spacecraft and the same throttleable thruster, with no impulsive burns.

Mission totals. Table 3 summarises the breakdown. The complete mission delivers $\Delta v_{\text{total}} \approx 2502 \text{ m s}^{-1}$ in 31.8 d of flight, with every metre per second produced by binary on/off commands of a single thruster.

Table 3: All-binary Earth–Moon mission breakdown.

Phase	Δv (m/s)	TOF (days)	Mode
1: Edelbaum spiral + injection boost	1924	6.4	main, tangential
2: cislunar correction (QUBO)	16	17.4	cruise, tangential
3: sliding-window QUBO capture	561	8.0	main, anti-tangential
Total	2502	31.8	all binary

Figure 4 shows the three-panel mission diagram. Phase 1 dominates the Δv budget at 1924 m s^{-1} (77% of the total), Phase 3 is the second-largest cost at 561 m s^{-1} (22%), and Phase 2 contributes 16 m s^{-1} (0.6%), essentially free in comparison; yet without the Phase-2 correction the 22 000-km miss precludes any usable perilune passage and the mission cannot continue.

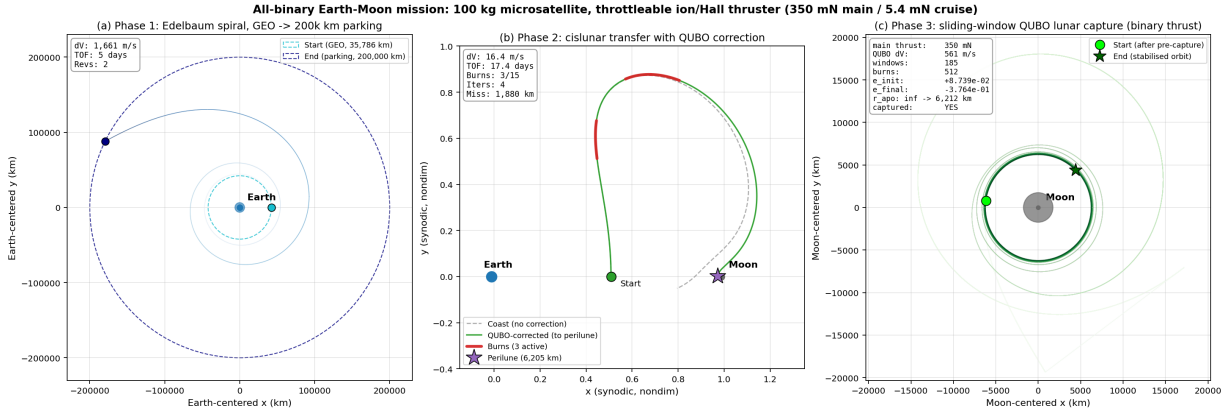


Figure 4: All-binary Earth–Moon mission overview. (a) Phase 1 Edelbaum spiral from GEO to a 200 000 km parking orbit, Earth-centered. (b) Phase 2 cislunar transfer under iterative-QUBO correction in the synodic frame; coast (grey dashed) drifts to a 22 000 km miss while the QUBO corrects to 1880 km miss with three burns. (c) Phase 3 sliding-window QUBO capture trajectory in the Moon-relative frame; the spacecraft enters hyperbolic from Phase 2 and is brought into a bound, near-circular orbit by binary anti-tangential pulses during ~ 200 short windows. The Δv breakdown and TOF totals are reported in Table 3.

The three-panel overview of Figure 4 plots each phase in the frame that is most natural for it (Earth-inertial for Phase 1, synodic for Phase 2 and 3), which can hide the fact that the trajectory is physically continuous. Figure 5 makes that continuity explicit.

Validation against flown missions. To place the all-binary budget of Table 3 in context, we compare it with three flown Earth–Moon missions spanning the propulsion regimes (Fig. 6). The apples-to-apples reference is ESA’s SMART-1, the canonical solar-electric low-thrust spiral to the Moon: it delivered $\approx 3.5 \text{ km s}^{-1}$ over ~ 13 months from a GTO start with a 70 mN Hall thruster [14, 15]. Our mission’s 2502 m s^{-1} sits in the same electric-propulsion band, and its propellant fraction at a comparable specific impulse ($I_{\text{sp}} \approx 1640 \text{ s}$) is $\approx 14\%$ against SMART-1’s $\approx 20\%$. Two caveats keep the comparison honest. First, our spiral departs a *circular* GEO whereas SMART-1 departed a GTO with a low perigee, which accounts for part of the lower Phase-1 cost. Second, and dominant, our assumed initial acceleration $a_0 = 3.5 \text{ mm s}^{-2}$ is roughly $18\times$ SMART-1’s 0.19 mm s^{-2} ; this — not any property of the scheduling — is why the present

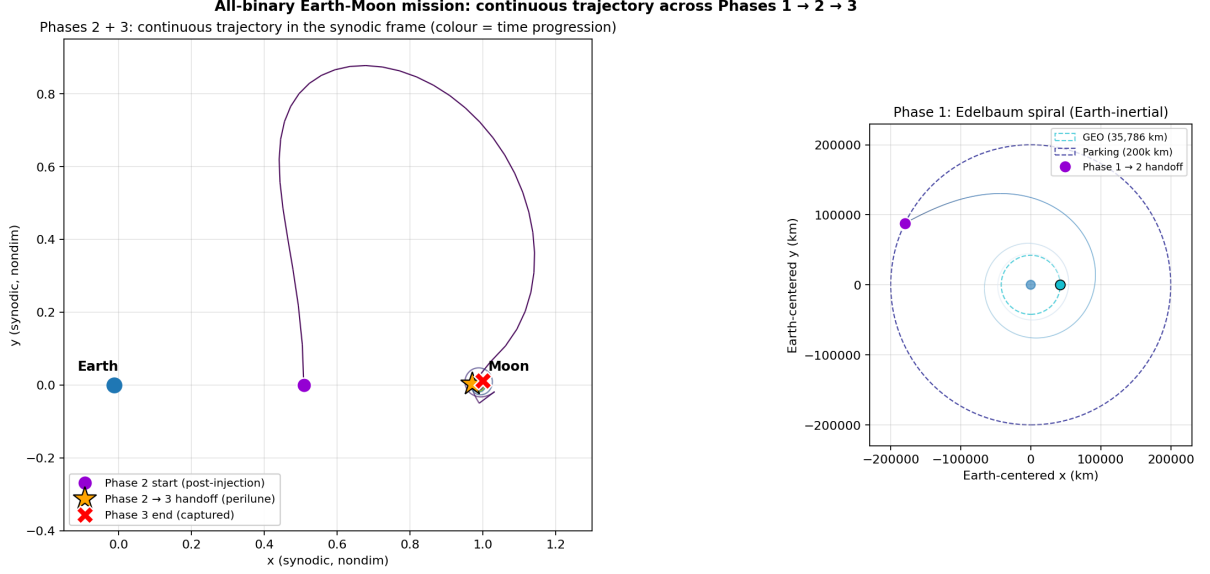


Figure 5: Continuity of the all-binary mission across Phases 1 → 2 → 3. *Left*: Phases 2 and 3 plotted as a single continuous trajectory in the synodic frame, with a viridis colour gradient encoding mission elapsed time; the orange star marks the Phase 2 → 3 handoff at perilune, which lies on the curve rather than separating two distinct curves. *Right*: Phase 1 in Earth-inertial coordinates with the GEO start and parking-orbit end circles; the handoff marker identifies the state that becomes the Phase-2 initial condition. The cumulative Δv and altitude histories are described in the surrounding text.

time of flight is ~ 1 month against SMART-1's ~ 13 , and it is an optimistic high-power assumption for a 100 kg bus.

To remove that confound entirely we re-run the *same* analytical Edelbaum spiral at SMART-1's own operating point: $a_0 = 0.19 \text{ mm s}^{-2}$, $I_{\text{sp}} = 1640 \text{ s}$, and a circular start orbit carrying SMART-1's GTO energy (semi-major axis $a_{\text{GTO}} \approx 24629 \text{ km}$). Because the Edelbaum Δv depends only on the start/end circular speeds and not on thrust, while the time of flight scales as $\Delta v/a_0$, this is a parameter-matched like-for-like check. The spiral to the Moon's sphere of influence then costs $\approx 2.9 \text{ km s}^{-1}$ over 136 revolutions, and at SMART-1's $\sim 40\%$ thrust duty cycle takes ≈ 440 days — against SMART-1's flown 3.5 km s^{-1} (which additionally includes lunar capture), 207 revolutions, and ~ 410 days. The agreement in Δv magnitude, revolution count, and duty-adjusted time of flight (the second qalunar marker in Fig. 6, which falls on SMART-1) confirms that the earlier ~ 1 -month figure reflected the $18\times$ higher thrust-to-mass of the demonstrator spacecraft, not any discrepancy in the transfer physics. The point is therefore not that the binary mission is cheaper, but that its end-to-end budget is physically consistent with flown electric-propulsion practice once the operating point is matched. CAPSTONE's low-energy ballistic transfer and Apollo's impulsive chemical transfer bracket the other two regimes in Fig. 6.

6 Discussion

When to use which QUBO formulation. The naturally-binary formulation is the natural fit when the engine is pulse-driven, when the mission constraints favour a finite list of discrete burn slots, or when the linearisation about a reference trajectory is faithful enough that the squared-norm objective is a good proxy for the true miss. For the Earth–Moon injection-correction scenario considered here, the linearisation around the unforced coast arc has three regimes that the experiments of Sec. 5 delineate.

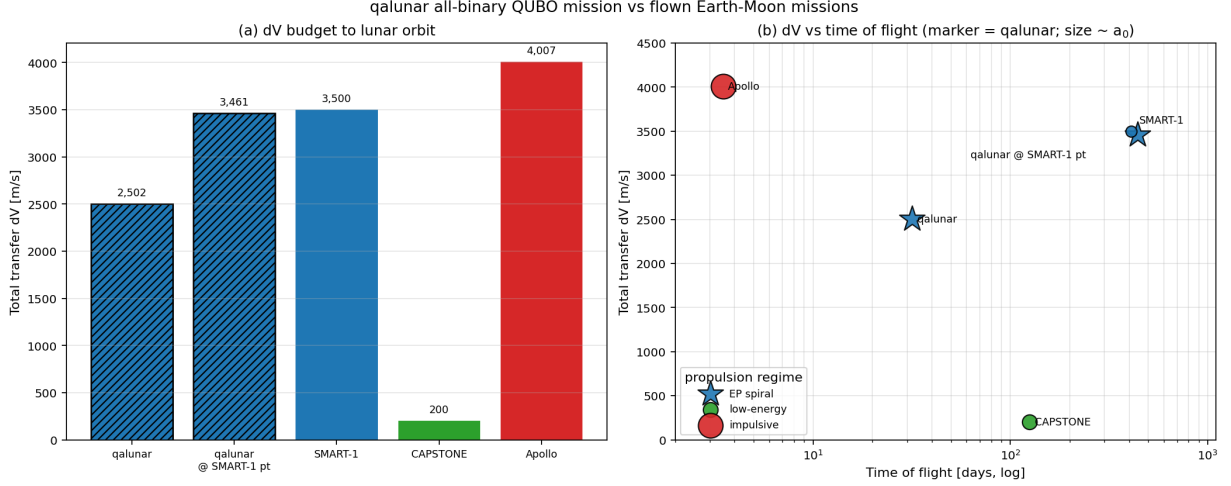


Figure 6: The all-binary mission against flown Earth–Moon missions. *Left*: total transfer Δv to lunar orbit; the hatched bars are this work. *Right*: Δv versus time of flight (log scale), coloured by propulsion regime and sized by initial thrust-to-mass a_0 . The two qalunar markers (stars) are the demonstrator end-to-end mission of Table 3 ($a_0 = 3.5 \text{ mm s}^{-2}$, ~ 1 month) and the *same* spiral re-run at SMART-1’s operating point ($a_0 = 0.19 \text{ mm s}^{-2}$, GTO-energy start, $\sim 40\%$ duty), which falls on SMART-1. Figures for SMART-1, CAPSTONE, and Apollo are published values (see text).

Trust-region rule as the operational difference. Single-pass QUBO solves are routinely reported in the QA literature without a trust-region accept rule, on the implicit assumption that the linearisation is faithful. Our experiments show this assumption holds only inside the single-pass-faithful regime ($T \lesssim 2.5$ nondim, ~ 11 days); beyond it, an unguarded single-pass can recommend a schedule that is worse than coast. The accept-or-reject rule of Algorithm 1 is the binary analogue of a trust-region method in continuous nonlinear optimisation [12]; we recommend that any QUBO-based trajectory tool adopt it as a default.

From single-arc QUBO to a complete binary mission. The end-to-end demo of Sec. 5.9 shows that the same scheduling formulation, applied window by window, extends from a single trajectory correction to an entire Earth–Moon transfer with no impulsive burns.

Comparison with state-of-the-art Earth–Moon transfers. Recent work by Taylor et al. [13] from the Purdue Orbital student team designs Earth-to-NRHO and Earth-to-LLO transfers via Pontryagin’s Minimum Principle and primer vector (PV) theory, with hyperbolic-tangent smoothing and homotopy continuation in ρ and $u_{\max}$ to converge on the impulsive limit. Their reported totals are 3.93 km s^{-1} in 5.2 d for LEO $\rightarrow$ NRHO and 3.91 km s^{-1} in 4.7 d for LEO $\rightarrow$ LLO. Table 4 places those numbers next to our end-to-end binary-QUBO result. The comparison is not apples-to-apples in three respects: (i) the starting orbit differs (GEO vs. LEO; a $\sim 3.9 \text{ km s}^{-1}$ Hohmann gap that favours qalunar), (ii) the propulsion system differs, and (iii) the final orbit differs (qalunar reaches a near-circular bound orbit at the perilune radius rather than 100 km LLO or the 9:2 NRHO). Subject to those caveats, the qualitative conclusion is that the binary-QUBO formulation produces a Δv budget consistent with the indirect-method state of the art at the cost of an order of magnitude longer TOF (the expected low-thrust trade-off). The methodological novelty here is not a Δv improvement but the demonstration that an entire Earth–Moon mission can be flown by binary on/off commands of a single throttleable thruster, with the QUBO scheduler and its sliding-window extension as the discrete-side analogue of the indirect primer-vector switching function [13].

Table 4: End-to-end Earth–Moon transfer comparison. The *Method* column lists the principal optimisation framework; totals are not directly comparable due to differences in starting orbit, propulsion system, and final orbit (see text).

Source	Transfer	Δv (km/s)	TOF (d)	Method
This work	GEO $\rightarrow$ bound lunar orbit	2.50	31.8	binary QUBO, low-thrust
[13]	LEO $\rightarrow$ 9:2 NRHO	3.93	5.2	PMP + PV, impulsive
[13]	LEO $\rightarrow$ LLO (100 km)	3.91	4.7	PMP + PV, impulsive

Limitations and future work. The present results use classical samplers as upstream proxies for quantum annealing. The extension to D-Wave Leap Hybrid (BQM and nonlinear) and to direct submission of the QUBO to the Advantage 2 quantum processor is a natural next step and is implemented behind a sampler interface in the accompanying code. Reporting hardware results is left to a follow-up. The single-channel formulation restricts the action set; a multi-channel run with simultaneous tangential, normal, and binormal axes is supported by the code but not yet swept parametrically. Beyond the planar problem, the same formulation extends to the full spatial CR3BP with an 8-state phase space and three thrust channels at the cost of a larger QUBO, but no qualitative changes to the methodology.

The all-binary mission of Sec. 5.9 reaches a near-circular bound orbit at the perilune radius (~ 4500 km altitude). A more interesting extension is a fully four-body model (Earth–Moon–Sun) that admits weak-stability-boundary trajectories in which the Moon-relative perilune speed approaches escape velocity naturally; in that regime the energy budget for binary capture collapses to a few metres per second and the throttling asymmetry between Phases 2 and 3 disappears. Finally, the sliding-window structure is currently parameter-agnostic: an adaptive choice of window length, action radius, and brake-factor based on the running orbital state would close the loop with an autonomous capture controller.

7 Conclusion

We presented a naturally-binary QUBO formulation of low-thrust cislunar trajectory scheduling in the planar CR3BP, complementary to the continuous-coefficient QUBO formulation of De Grossi, Carbone et al. The binary decision variables avoid the quantisation floor that plagues the continuous formulation, and the per-step on/off semantics matches both the physical engine operation and the quantum-annealing variable space. A sequential re-linearisation outer loop with a trust-region accept rule closes the linearisation gap on long arcs and abstains outside its basin of attraction, properties the single-pass solve does not have. Classical baselines via McCormick-lifted MILP certify optimality up to $N \approx 20$, and the SA-vs-MILP scaling crossover near $N \approx 25$ – 30 delineates the regime where moving to D-Wave hardware becomes quantitatively defensible.

A sliding-window extension of the iterative driver chains short single-arc QUBOs across many lunar revolutions and, paired with an action-radius gate and a velocity-only circular target, captures a hyperbolic flyby into a bound near-circular orbit under purely binary anti-tangential thrust. We use it together with the original cislunar-correction QUBO to demonstrate an end-to-end Earth–Moon mission in which every metre per second is delivered by on/off commands of a single throttleable ion/Hall thruster, with no impulsive burns. The mission delivers $\Delta v_{\text{total}} \approx 2502 \text{ m s}^{-1}$ in ~ 32 d and shows that the QUBO scheduling formulation, originally proposed for single-arc trajectory correction, composes into a complete mission profile when its operating regime is respected by appropriate throttling and action-region gating.

Acknowledgements

This work was carried out at the School of Aerospace Engineering, Sapienza Università di Roma, under the supervision of Prof. Antonio Paolozzi.

References

- [1] De Grossi, F., Carbone, A., Spiller, D., Ottaviani, D., Mengoni, R., Circi, C., *Transcription and optimization of an interplanetary trajectory through quantum annealing*, *Astrodynamics* **9**:195–215, 2025.
- [2] Szebehely, V., *Theory of Orbits: The Restricted Problem of Three Bodies*, Academic Press, 1967.
- [3] Koon, W. S., Lo, M. W., Marsden, J. E., Ross, S. D., *Dynamical Systems, the Three-Body Problem and Space Mission Design*, Springer, 2011.
- [4] Kelly, M., *An introduction to trajectory optimization: How to do your own direct collocation*, *SIAM Review* **59**(4):849–904, 2017.
- [5] Betts, J. T., *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming*, 2nd ed., SIAM, 2010.
- [6] Edelbaum, T. N., *Propulsion Requirements for Controllable Satellites*, *ARS Journal* **31**(8):1079–1089, 1961.
- [7] Glover, F., Kochenberger, G., Du, Y., *A tutorial on formulating and using QUBO models*, *4OR* **17**(4):335–371, 2019.
- [8] Albash, T., Lidar, D. A., *Adiabatic quantum computation*, *Reviews of Modern Physics* **90**:015002, 2018.
- [9] McCormick, G. P., *Computability of global solutions to factorable nonconvex programs: Part I – Convex underestimating problems*, *Mathematical Programming* **10**(1):147–175, 1976.
- [10] Huangfu, Q., Hall, J. A. J., *Parallelizing the dual revised simplex method*, *Mathematical Programming Computation* **10**(1):119–142, 2018.
- [11] D-Wave Systems Inc., *D-Wave System Documentation: Solver Computation Time*, ocean.dwavesys.com, 2024.
- [12] Conn, A. R., Gould, N. I. M., Toint, P. L., *Trust-Region Methods*, MPS-SIAM Series on Optimization, SIAM, 2000.
- [13] Taylor, T. E., Madden, K. F., Bohne, S. L., Vetter, M. J., *An Earth–Moon Trajectory and Mission Design*, Purdue Orbital, Purdue University, West Lafayette, IN, 2026.
- [14] Racca, G. D., et al., *SMART-1 mission description and development status*, *Planetary and Space Science* **50**(14–15):1323–1337, 2002.
- [15] Milligan, D., Camino, O., Gestal, D., *SMART-1 electric propulsion operations*, in *41st AIAA/ASME/SAE/ASEE Joint Propulsion Conference*, AIAA 2005-3671, Tucson, AZ, 2005.